AP CALCULUS AB

Unit 1 • Self Check 1.1 and 1.2 • Lessons 1-1 to 1-4 • Answer key

Self Check 1.1 and 1.2 • Lessons 1-1 to 1-4

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

$$\lim_{x\to -3} \frac{x^2+x-6}{x^2-9} =$$

(A) $\frac{5}{6}$

(B) 0

(C) $-\frac{5}{6}$

(D) It does not exist.

Solution

Answer: (A). $\frac{(x+3)(x-2)}{(x+3)(x-3)} = \frac{x-2}{x-3} \to \frac{-5}{-6}$.

Question 2

NO CALCULATOR

$$\lim_{x\to \infty} \frac{2x-x^3}{4x^3+1} =$$

(A) 2

(B) $-\frac{1}{4}$

(C) 0

(D) It does not exist.

Solution

Answer: (B). Same degree: $\frac{-1}{4}$.

Question 3

NO CALCULATOR

Let $j(x) = \begin{cases} x+3, & x<1 \\ 2x, & x \geq 1. \end{cases}$ Then $\lim_{x\to 1} j(x) =$

(A) 2

(B) 4

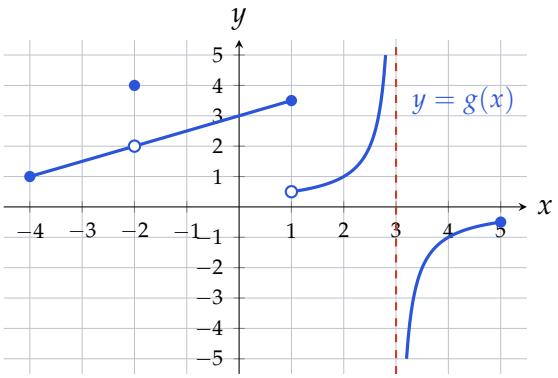
(C) 3

(D) It does not exist.

Solution

Answer: (D). Left 4, right 2.

Question 4 and the free response refer to the graph of g.



Question 4

NO CALCULATOR

Using the graph of g, at which x is the discontinuity removable?

(A) $x = -2$
(B) $x = 1$
(C) $x = 3$
(D) $x = -4$

Solution
Answer: (A). The limit at $x = -2$ is 2, but $g(-2) = 4$.

Question 5 (Free response)

NO CALCULATOR

Using the graph of g, classify the discontinuities at $x = -2$, 1 and 3, with limit statements.

Solution
-2: removable ($\lim = 2 \neq 4 = g(-2)$). 1: jump (left 3.5, right 0.5). 3: infinite ($+\infty$ from the left, $-\infty$ from the right).